Soungchul Cha, Ph.D.

soungchul@gmail.com
linkedin.com/in/soungchul-cha-b0252a32
https://soungchulcha.github.io/portfolio/index.html

Principal Scientist Profile

Innovative immunologist and translational scientist with 20+ years of experience in cancer vaccine discovery and development. Expertise in neoantigen identification, T-cell immunology, and in-vitro immune assay design supporting next-generation vaccine and immunotherapy platforms. Proven leadership in bench-to-clinic translation of DNA and idiotype-based vaccines, advancing multiple candidates through Phase I/II clinical trials. Skilled in mechanistic immunology, in-vitro assay development, and collaborative cross-functional research driving IND-enabling studies.

Core Strengths

- Personalized and tumor-specific vaccine development
- Antigen discovery, neoepitope validation, and immunogenicity screening
- T-cell receptor (TCR) discovery and immune-monitoring platform design
- Translational biomarker strategy and mechanistic immunology
- Cross-functional collaboration with bioinformatics, GMP manufacturing, and clinical teams

Selected Achievements

- Developed LLM-enabled multi-omics workflows integrating scRNA-seq, WES, bulk RNA-seq, and long-read RNA-seq, empowering non-specialists to perform independent exploratory analyses and reducing reliance on dedicated bioinformatics support.
- Advanced personalized cancer vaccines from preclinical models to Phase I/II trials at City of Hope and MD Anderson, establishing translational workflows from neoantigen prediction to immune monitoring.
- Developed an in-vitro immune-monitoring platform using autologous B cells transfected with antigen mRNA—enabling detection of low-frequency, antigen-specific T cells in PBMC and marrow samples.
- Designed and validated T-cell polyfunctionality assays for CAR-T and vaccine-induced responses, predicting clinical benefit and mechanistic biomarkers.
- Built a neoantigen discovery pipeline integrating tumor sequencing, bioinformatics ranking (IC50, percentile rank), and experimental validation.
- Led cross-disciplinary immunology teams supporting IND-enabling and translational vaccine programs of DNA vaccine formulations

Professional Experience

1. **Staff Scientist, City of Hope National Medical Center,** Duarte, California, United States, Nov 2017 – Present

- Developed and implemented LLM-enabled multi-omics analysis workflows integrating scRNA-seq, WES, bulk RNA-seq, and long-read RNA-seq to support independent exploratory analyses by experimental researchers and reduce reliance on dedicated bioinformatics resources.
- Lead and manage translational research programs focused on cancer vaccine discovery and immune-monitoring platform development.
- Directed the immunology monitoring team for a randomized Phase II clinical trial evaluating idiotype vaccination combined with adoptive autologous T-cell transfer in multiple myeloma patients.
- Developed and optimized a highly sensitive immune-monitoring assay capable of detecting low-frequency, antigen-specific T cells in PBMC and bone marrow samples from vaccinated patients.
- Established a novel in-vitro immune-monitoring platform using autologous immortalized B cells transfected with antigen-encoding mRNA to discover new antigen-specific T-cell receptors (TCRs).

- Designed and validated assays assessing polyfunctional cytokine responses of BAFFR-CAR-T cells, correlating polyfunctionality with predicted clinical outcomes.
- Collaborated cross-functionally with the Genomics Core, Bioinformatics, and Wet Lab teams to characterize tumor microenvironment alterations before and after vaccination.
- Built a neoantigen discovery and validation pipeline integrating genomic analysis with experimental validation to identify patient-specific targets for vaccine design.
- Mentor and supervise junior scientists and postdoctoral fellows, fostering skill development and scientific leadership within the immunology program.
- Lead translational development of a novel TLR7/8-based personal cancer vaccine, advancing preclinical findings toward clinical evaluation.

2. Assistant Professor, MD Anderson Cancer Center, Houston, United States, May 2008 - Nov 2017.

- Led translational research initiatives advancing a personalized DNA vaccine program from concept through Phase I clinical trials, collaborating closely with manufacturing, clinical, and regulatory teams to ensure scientific rigor and compliance.
- Developed and implemented Standard Operating Procedures (SOPs) for idiotype generation, encompassing antigen processing, plasmid construction, and comprehensive quality control.
- Established a dedicated DNA vaccine laboratory, enabling in-house production of idiotype plasmid constructs directly from patient-derived tumor B cells under GMP-aligned processes.
- Identified lymphoma-specific antigens from tumor biopsies and successfully generated clinical-grade DNA vaccines, facilitating the first-in-human vaccination studies at MD Anderson.
- Contributed to cross-departmental translational programs integrating immunology, genomics, and clinical development to accelerate bench-to-clinic vaccine translation.

3. Instructor, MD Anderson Cancer Center, Houston, United States, May 2004 - May 2008.

- Developed and optimized vaccine adjuvants and delivery systems in murine tumor models to enhance antigen presentation and T-cell priming efficiency.
- Established a neoantigen identification workflow for DNA vaccine development using lymphoma patient tumor samples, integrating molecular cloning and immunogenicity validation.
- Pioneered antigen cloning techniques by generating and screening a modified phage-displayed cDNA expression library with patient-derived sera to isolate B-cell lymphoma–associated antigens.
- Supported translational immunology programs that bridged preclinical antigen discovery with clinical vaccine development efforts.

4. Industry Consultantships and Scientific Advisory Board

- PEPROMENE Bio, Irvine, CA, Consultant, 3/2020-present.
- RENHAIM THERAPEUTICS. Gyeonggi-do, Korea, Scientific Advisory Board, 7/2024 – 7/2025
- PROGENEER. Seoul, Korea, Scientific Advisory Board, 7/2025 – present.

5. Patents and Technology Licenses

Patents

- WO0215048748 A1: MYELOID-DERIVED SUPPRESSOR CELL-SPECIFIC PEPTIDES FOR DIAGNOSTIC AND THERAPEUTIC USE. September 30, 2013

Education

Doctor of Philosophy, Microbiology and Immunology 1992-1999

Konkuk University, Seoul, Republic of Korea

Postdoctoral Research Fellowship 1999-2004

National Institutes of Health/ National Cancer Center, Experimental Immunology Branch, Frederick, MD